
ME 595: Experiential Learning in Composite Design, Manufacturing, Testing and Recycling

Course Code: ME 595

Course Name: Experiential Learning in Composite Design, Manufacturing, Testing and Recycling

L-T-P-C: 1-0-4-3

Prerequisites: Consent of Faculty Advisor

Intended for: UG/M.Tech/PhD

Mutual Exclusion: None

Approval: 61st BoA

Preamble

Polymer Matrix Composites (PMCs) are advanced materials widely used in aerospace, automotive, marine, renewable energy, and defense sectors due to their high strength-to-weight ratio, corrosion resistance, and design flexibility. This course offers a holistic experiential learning framework that spans composite design, multi-scale simulation (micro–meso–macro), hand lay-up, vacuum-assisted resin transfer molding (VARTM) for manufacturing, and standardized mechanical testing to evaluate tensile, compressive, flexural, and impact performance. Students will also gain hands-on expertise in advanced characterization methods such as Scanning Electron Microscopy (SEM) to study fracture morphology and failure modes, including delamination, fatigue delamination, and matrix or fiber cracking. A distinctive feature of the course is its focus on sustainability, with modules on recycling technologies, fiber recovery from waste composites, and circular economy practices to address environmental challenges. This course provides a comprehensive framework for the design, simulation, manufacturing, testing, and recycling of polymer matrix composites (PMCs).

Course Objectives

- To introduce fundamental principles of polymer matrix composites and their design methodologies.
- To impart skills in micro–meso–macro simulation strategies for predicting composite performance.
- To provide practical training in hand lay-up and VARTM-based manufacturing.
- To develop competency in standard mechanical testing of composites.
- To analyze fracture morphology and modes through SEM and other advanced characterization techniques.
- To familiarize students with sustainability challenges, recycling technologies, and fiber recovery from composite waste.
- To cultivate awareness of circular economy principles and environmental effects of composites.

Course Modules with quantitative lecture hours

- **Practical 1: Composite Fabrication (Hand lay-up and VARTM Hands-on):** (6 hrs)
 - Introduction and Methodologies for composite materials
 - Students prepare molds, perform resin infusion under vacuum, cure laminates, and evaluate quality (voids, fiber wet-out).
 - Exposure to real manufacturing challenges such as leakage, incomplete impregnation, or resin flow control.

- **Practical 2: Multi-scale Simulation Projects:** (8 hrs)
 - Micro-macro mechanics introduction, and their importance, case studies of RVE based design and classical laminate theory
 - Micro-scale: Building a Representative Volume Element (RVE) and predicting fiber/matrix properties.
 - Meso-scale: Laminate design and delamination modeling.
 - Macro-scale: Finite Element Analysis (FEA) of a composite beam or panel under impact load.
- **Practical 3: Mechanical Testing Laboratory:** (6 hrs)
 - Mechanical Properties of composite materials and their ASTM standards
 - Conduct tensile, compression, bending, and impact tests as per ASTM/ISO standards.
 - Collect and interpret load–displacement and stress–strain data.
 - Compare experimental results with simulation predictions.
- **Practical 4: Fracture Morphology Study:** (4 hrs)
 - Identification of fiber pull-out, matrix cracking, interfacial debonding, and delamination.
 - Correlation of fracture features with failure modes.
- **Practical 5: Recycling & Fiber Recovery:** (6 hrs)
 - Small-scale demonstration of mechanical or thermal recycling of cured composites.
 - Fiber extraction trials (carbon/glass fibers).
 - Property evaluation of recycled fibers compared to virgin fibers.
- **Practical 6: Circular Economy Case Study:** (6 hrs)
 - Group project analyzing the lifecycle of composites in aerospace/automotive/wind energy sectors.
 - Proposal of sustainable design and recycling solutions for end-of-life composites.
- **Practical 7: Capstone Project: Design → Manufacture → Test → Recycle Cycle:** (24 hrs)
 - Teams design a composite component (e.g., plate, beam, or panel).
 - Manufacture it via Hand lay up or VARTM.
 - Test its mechanical performance.
 - Computational simulation for performance/properties evaluation
 - Analyze fracture morphology.
 - Attempt partial recycling and assess recovered materials.

Project-Based Problems for Experiential Learning

- **A. Design & Simulation (Micro–Meso–Macro):**
 - Develop a Representative Volume Element (RVE) to predict elastic constants of a unidirectional composite.
 - Model stress distribution at the fiber–matrix interface under transverse loading.
 - Simulate the effect of fiber misalignment on tensile strength.

- Study impact of fiber waviness on stiffness using finite element analysis.
- Analyze progressive matrix cracking in a cross-ply laminate under tensile load.
- Model delamination propagation using cohesive zone elements.
- Conduct multi-scale simulation to connect micro-level properties with structural performance.
- Optimize stacking sequence of a laminate for maximum bending stiffness.
- Study the influence of fiber volume fraction on overall composite modulus.
- Simulate drop-weight impact on a composite plate and validate with experiments.
- Investigate fatigue life of a composite plate under cyclic loading.
- Study vibration response of a composite beam using FEA and compare with experiment.

• **B. Manufacturing & Processing:**

- Fabricate glass fiber/epoxy laminates using Hand layup method and VARTM; and measure void fraction.
- Compare properties of laminates fabricated at different vacuum levels.
- Study effect of resin viscosity on resin infusion time.
- Manufacture laminates with different curing cycles and evaluate mechanical properties.
- Investigate influence of mold release agents on surface finish and quality.
- Study effect of fiber stacking sequence on resin infusion rate.
- Develop hybrid laminates (carbon + glass fibers) and evaluate performance.
- Compare laminates fabricated with and without peel ply layers.
- Measure permeability of different fiber preforms for VARTM.
- Conduct flow visualization experiments to analyze resin front progression.
- Optimize mold design for uniform resin distribution.
- Study influence of compaction pressure on fiber volume fraction in laminates.

• **C. Mechanical Testing & Performance Evaluation:**

- Perform tensile testing of fabricated laminates and compare with theoretical predictions.
- Conduct flexural tests (three-point and four-point bending) and analyze failure modes.
- Evaluate compressive strength of composites using short-block method.
- Carry out interlaminar shear strength (ILSS) testing.
- Perform impact testing (drop-weight or Charpy/Izod) on composite panels.
- Study effect of laminate thickness on impact energy absorption.
- Compare mechanical performance of unidirectional vs. woven laminates.
- Analyze effect of fiber orientation (0° , 45° , 90°) on tensile/flexural strength.
- Study effect of repeated low-velocity impacts on composite durability.
- Evaluate mechanical property degradation after thermal aging cycles.
- Measure fatigue life of laminates under tension–tension cyclic loading.
- Study creep behavior of polymer composites at elevated temperatures.

• **D. Fracture Morphology & Failure Modes:**

- Use SEM to analyze fracture surfaces of tensile-tested samples.
- Compare fracture morphology of composites with brittle vs. toughened matrices.
- Identify fiber pull-out, matrix cracking, and interfacial debonding in flexural failure.
- Study fracture surfaces from impact tests and correlate with absorbed energy.
- Analyze delamination regions in cross-ply laminates after cyclic loading.
- Compare SEM images of virgin and thermally aged composites to study degradation.
- Investigate effect of stacking sequence on fracture morphology.
- Correlate fracture features with predicted failure modes from simulation.

• **E. Recycling, Fiber Recovery & Circular Economy:**

- Demonstrate mechanical recycling of waste composites via grinding and reuse in secondary products.
- Extract glass fibers from end-of-life composites using pyrolysis and analyze fiber quality.
- Recover carbon fibers using chemical recycling methods and compare properties with virgin fibers.
- Manufacture hybrid laminates using a mix of virgin and recycled fibers and test mechanical properties.
- Evaluate thermal stability of recycled fibers using DSC/TGA.
- Conduct a lifecycle assessment (LCA) of a composite product (e.g., wind turbine blade, automotive panel) and propose sustainable recycling strategies.

Text Books

1. P.K. Mallick – *Fiber-Reinforced Composites: Materials, Manufacturing, and Design* (3rd Ed., CRC Press, 2007)
2. Robert M. Jones – *Mechanics of Composite Materials* (Taylor & Francis, 1999).

References

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2. A.B. Strong – *Fundamentals of Composites Manufacturing: Materials, Methods, and Applications* (2nd Ed., SME, 2008)
3. S.T. Peters (Ed.) – *Handbook of Composites* (2nd Ed., Chapman & Hall, 1998)
4. S. Gay, J. Hoa & S.W. Tsai – *Composite Materials: Design and Applications* (3rd Ed., CRC Press, 2015)
5. L. Tong, A.P. Mouritz & M. Bannister – *3D Fibre Reinforced Polymer Composites* (Elsevier, 2002)
6. S.J. Pickering (Ed.) – *Recycling Technologies for Thermoset Composites* (Elsevier, 2016)
7. Y. Yang, R. Boom, B. Irion et al. – *Recycling of Composite Materials* (Springer, 2012)
8. M. Piggott – *Load Bearing Fibre Composites* (2nd Ed., Springer, 2002)
9. F.L. Mathews and R.D. Rawlings – *Composite Materials: Engineering and Science* (Elsevier, 1999)
10. I.M. Daniel & O. Ishai – *Engineering Mechanics of Composite Materials* (2nd Ed., Oxford University Press, 2006)